

BRANDON M. DURAN

PhD Candidate, Scripps Institution of Oceanography

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RESEARCH INTERESTS

aerosol-cloud interactions, aerosol radiative forcing, circulation responses to aerosol forcing, satellite simulators

EDUCATION

Scripps Institution of Oceanography, University of California San Diego 2022 - Present

Masters in Earth Science (2024), Ph.D. expected in 2027

Advisors: Profs. Nicholas Lutsko & Shang-Ping Xie

Committee Members: Profs. Duncan Watson-Parris, Casey Wall, & Amato Evan

Georgetown University

2018-2022

BA in Physics, Computer Sciences

PEER REVIEWED PUBLICATIONS

4. **Duran, B.M.**, Lutsko, N.J., and Wall, C.J. Aerosol-Ice-Cloud Interactions in a Perturbed Parameter Ensemble (2026). *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-25-0724.1>.
3. Michibata, T., Wall, C.J., Hirota, N., **Duran, B.M.**, and Nozawa, T. Recent Advances in the Observation and Modeling of Aerosol-Cloud Interactions, Cloud Feedbacks, and Earth's Energy Imbalance: A Review (2025). *Current Pollution Reports*. <https://doi.org/10.1007/s40726-025-00382-6>.
2. Wall, C.J., Paynter, D., Qin, Y., Debolskiy, M., Duffy, M.L., Michibata, T., **Duran, B.M.**, Lutsko, N.J., Ma, P.L., Medeiros, B., Storelvmo, T., and Zhao, M. Decomposing Cloud Radiative Feedbacks by Cloud-Top Phase (2025). *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-24-0538.1>.
1. **Duran, B.M.**, Wall, C.J., Lutsko, N.J., Michibata, T., Ma, P.L., Qin, Y., Duffy, M.L., Medeiros, B., and Debolskiy, M. A new method for diagnosing effective radiative forcing from aerosol-cloud interactions in climate models (2025). *Atmospheric Chemistry and Physics*. <https://doi.org/10.5194/acp-25-2123-2025>.

SUBMITTED MANUSCRIPTS

2. **Duran, B.M.**, Wall, C.J., Lutsko, N.J., Wynn, S., Watson-Parris, D., and Mülmenstädt, C.J. A new proxy for the LWP adjustment from first principles (preprint, 2026) *ESS Open Archive*. <https://doi.org/10.22541/essoar.15006786/v1>
1. Wynn, S., Watson-Parris, D., **Duran, B.M.**, and Mülmenstädt, C.J. Base-state dependence of parameter sensitivity for the Twomey effect in a perturbed parameter ensemble (preprint, 2026) *Atmospheric Chemistry and Physics*. <https://doi.org/10.5194/egusphere-2026-2811>

AWARDS RECEIVED

2022	National Science Foundation Graduate Research Fellow
2020-2022	National Hispanic Scholar
2021	AGU Student Travel Grant Recipient

CONFERENCE PRESENTATIONS

- 2026** Oral: 2026 CESM Workshop
- 2026** Oral: Weather & Climate Livestream
- 2026** Oral: CESM Winter Working Group Meeting
- 2025** Poster: Gordon Research Conference, Radiation & Climate at Bates College
- 2025** Oral: 2025 CESM Workshop
- 2025** Invited Talk: Equilibrium Climate Sensitivity and Cloud Feedback Virtual Symposium
- 2024** Poster: AGU, Washington, D.C.
- 2024** Poster: Micro2Macro Workshop, University of Wyoming
- 2024** Oral: CFMIP Conference at Boston University

TEACHING AND MENTORING

Teaching Assistant

- 2026 *Dynamics of Atmos & Climate*, UCSD. Instructor: Nicholas Lutsko.
- 2025 *The Atmosphere*, UCSD. Instructor: Lynn Russell.
- 2021 *Mechanics and Electromagnetic Phenomena*, Georgetown University.
Instructor: Patrick Johnson.

Guest Lecturer

- 2026 *Dynamics of Atmos & Climate*, UCSD. Instructor: Nicholas Lutsko.

Research Mentor

- Sophie Wynn, PhD, SIO
- Devan Velji, undergraduate, UCSD

Peer Mentor

- Wenjing Zhang, SIO
- David Cera, SIO

SERVICE

Department

- 2026-** Student Transportation Advisory Committee
- 2024-** Co-organizer of SIO Climate Journal Club
- 2024-26** Climate Sciences representative and senior co-chair of the
Scripps Graduate Student Council
- 2024-25** SIO Extreme Weather and Lecturer faculty search committees
- 2023-24** Co-organizer of the Climate, Atmospheric Sciences and Physical
Oceanography (CASPO) seminar series

Other

- 2024-** Reviewer for *Science Advances*, *JGR: Atmospheres*, *Nature Climate Change*, and
Atmospheric Chemistry and Physics

PROFESSIONAL REFERENCES

- Nicholas Lutsko**, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: nlutsko@ucsd.edu
- Shang-Ping Xie**, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: sxie@ucsd.edu
- Duncan Watson-Parris**, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: dwatsonparris@ucsd.edu
- Sloan Coats**, Department of Earth Sciences, University of Hawai'i at Mānoa, Honolulu, HI, USA: scoats@hawaii.edu